

Chiral Water Structures on the Dolomite(104)-S Surface Enantiomer

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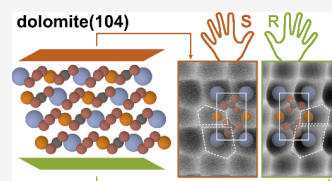
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Supporting Information

ABSTRACT: The particular distribution of dolomite over geological time scales and the challenges of growing dolomite under geological conditions in a laboratory setting are still some of the most intriguing scientific puzzles. So-called organogenic models that suggest the role of molecules, especially of extracellular polymeric substances in the liquid phase, in dolomite formation currently challenge the attempts to understand dolomite formation in the context of kinetic constraints. Here, we confirm the presence of chirality on dolomite(104) surfaces by atomic-scale imaging with noncontact atomic force microscopy. Two enantiomers are mapped at the two opposing crystal surfaces. Chiral signatures are further identified in the adsorption of single water molecules, and chirality transfer to a full water monolayer is imaged. Links between chiral specificity of dolomite growth and chiral selectivity of biomolecules on earth, mediated by the liquid phase, appear to be a so far overlooked jigsaw piece in understanding the formation processes of dolomite.

KEYWORDS: dolomite, water, chirality, noncontact atomic force microscopy



Dolomite ($\text{CaMg}(\text{CO}_3)_2$) is among the most intriguing minerals on earth.¹ The particular distribution of dolomite over geological time scales and the failures of growing ordered dolomite in a laboratory setting under geological conditions highlight the lack of understanding of the involved processes in this so-called “dolomite-problem”.² In particular, dolomite is significantly more prevalent in ancient sedimentary rock formations than in Holocene sediments^{3–5} and it hardly precipitates from the modern ocean despite a dolomite supersaturation in seawater.^{3,6} The latter observation goes along with the challenges of laboratory-based synthesis of dolomite under geological conditions: Running reactions even as long as 32 years at room temperature failed to form ordered dolomite;⁶ laboratory-based precipitation has so far only been realized at elevated temperatures (150–300 °C).^{7,8}

The term dolomite describes a rock-forming mineral series within the class of carbonates that encompasses varying chemical composition, order, and lattice structure.² The dolomite problem concerns the fabrication of ordered dolomite where the Mg:Ca ratio is near to 1:1 and the cations are ordered (i.e., alternating along [001]) within the crystal lattice structure. Reports on disordered dolomite or proto-dolomite are steps along the path of researching ordered dolomite crystal growth and even quantifying order in experiments can be a challenge.⁹ While kinetic factors set by constraints of the geological environment have intensively been discussed as factors in controlling and steering dolomite formation,² microbe-facilitated mechanisms i.e. as in organogenic models, might offer alternative explanations.¹⁰ In particular, the decomposition of organic matter into extracellular polymeric substances in the surrounding solution has been suggested to generate favorable environments for dolomitization,¹¹ with mechanisms involving elusive stabiliza-

tion capabilities of cations by negatively charged chemical moieties stemming from larger biomolecules.¹⁰ Yet, a consensus has been reached that mineral formation processes involve hydrated layers;² the formation of dolomite and many other minerals require the presence of fluid phases.¹² Most interestingly, chiral water structures have been mapped around chiral biomolecules.^{13–15} Thus, structural aspects on generating the particular dolomite cation order in the presence of chirality, imprinted on the liquid phase by chiral biomolecules, appear to be an overlooked aspect.

In this paper, we deliver compelling evidence for the presence of chiral (104) faces of dolomite, a property following from the bulk crystal structure, and image chirality transfer to the first water monolayer. Conclusions are drawn based on noncontact atomic force microscopy (NC-AFM) data acquired with CO-functionalized tips, partnered with density functional theory (DFT) calculations and image simulations. Strong molecule–surface interactions are revealed that mediate the chirality transfer. Our results deliver a new angle on the dolomite problem and disclose chirality as a so far hidden variable in the processes that concern ordered dolomite. Based on the mapping of chiral water structures around chiral biomolecules,^{13–15} our findings further suggest the function of water as a transducer from chiral selectivity of biomolecules to dolomite formation.

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The {104} faces resemble the most stable cleavage planes of dolomite, a common property also for the related carbonate minerals calcite (CaCO_3) and magnesite (MgCO_3).¹⁶ The rectangular (1×1) surface unit cell has dimensions of $0.771 \text{ nm} \times 0.481 \text{ nm}$ ^{17,18} and contains one Mg^{2+} ion, one Ca^{2+} ion, and two carbonate CO_3^{2-} groups. The carbonate groups are rotated within the surface plane such that one oxygen atom (O_{top}) of the carbonate group protrudes from the surface, O_{middle} lies mostly within the Mg–Ca plane, and O_{bottom} is located below the Mg–Ca plane, see side view in Figure 1(a). In fact, the dolomite crystal structure (crystallographic space group $R\bar{3}19$) dictates the presence of two enantiomeric (104) and ($\bar{1}0\bar{4}$) faces on two opposite crystal sites,²⁰ see Figure 1(a).

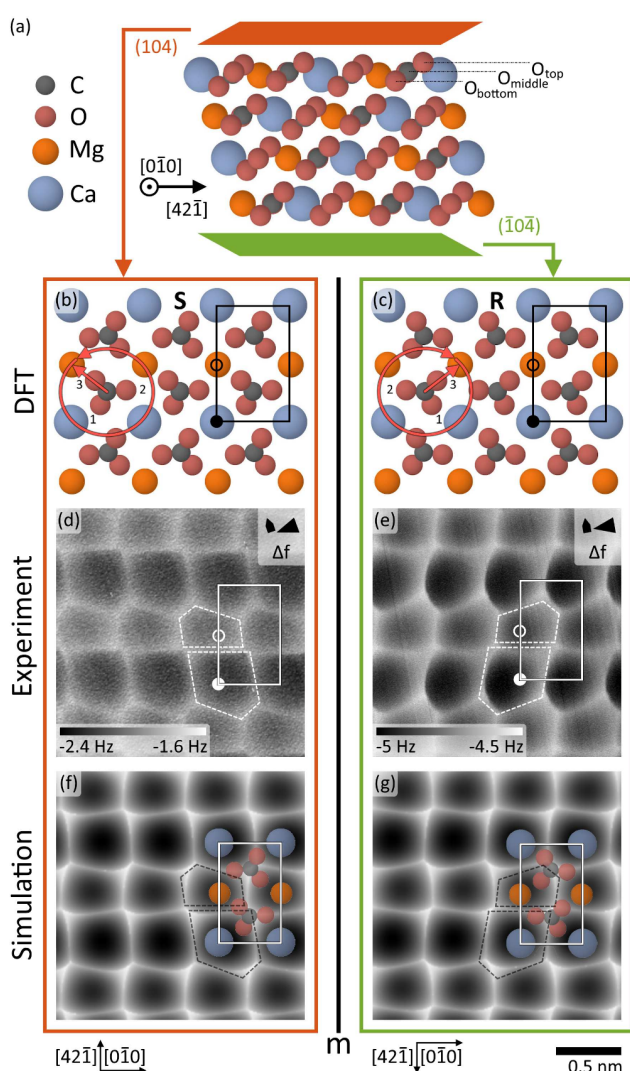


Figure 1. (a) Dolomite bulk crystal structure with two enantiomeric (104)^S and (104)^R surface planes. (b,c) DFT-optimized geometry of the dolomite(104)^S and (104)^R surfaces (unit cell marked by black rectangle; priorities of 1 to 3 around chiral carbon centers assigned to carbonate oxygens; enantiomeric assignment using the CIP rule visualized by red arrows). (d,e) Experimental high-resolution frequency shift data obtained with CO-functionalized tips on two opposing dolomite crystal surfaces and (f,g) corresponding simulated images, extracted at $z = 830 \text{ pm}$ (z is measured relative to the average topmost Ca plane). Unit cells marked by white rectangles.

To confirm this indication, we perform DFT geometry optimizations (see Supporting Information (SI) section A for details on the methods) to find the optimal structures of both the (104) and ($\bar{1}0\bar{4}$) surfaces. In fact, the structures in Figure 1(b,c) represent the optimized geometries of the two enantiomeric surfaces resulting from independent DFT runs, whereby the surface species only undergo minor relaxations from the bulk-truncated positions (Mg/Ca less than 1 pm and surface O maximum relaxations around 50 pm).

In analogy to molecular chirality, we assign the chiral centers on these surfaces to the carbon atoms. By adapting the Cahn-Ingold-Prelog rule (see also SI section B), we allocate priorities to the three carbonate oxygens based on the atomic numbers Z ($Z_{\text{Mg}} = 12$, $Z_{\text{Ca}} = 20$) of the neighboring surface ions, resulting in the highest (lowest) priority for the oxygen O_{bottom} (O_{top}). Therefore, we find a counterclockwise rotation of the arrangement of the species of the (104) plane shown in Figure 1(b), which is reversed in a clockwise manner for the ($\bar{1}0\bar{4}$) surface as indicated in Figure 1(c). Consequently, we refer to the (104) surface as enantiomer (104)^S and to the ($\bar{1}0\bar{4}$) surface as enantiomer (104)^R.

While the mirror symmetry between these two models is already apparent visually, the chiral property of these two structures is confirmed within DFT accuracy by a numerical symmetry analysis, see SI section C. As a rule of thumb, the position of the topmost oxygen atom O_{top} next to the magnesium cation is either left (enantiomer S, Figure 1(b)) or right (enantiomer R, Figure 1(c)) of the central carbon atom (see also red straight arrows in Figure 1(b,c)). This chiral submotif is mirror-symmetric to each other; yet, it cannot be superimposed on each other without reflection. Note that the unit cells are positioned in such a way that the mirror symmetry is directly apparent. With this choice, the $[42\bar{1}]$ direction proceeds in opposite directions between the two cases, highlighting that a careful assignment of the surface directions is required as a direct result of the presence of the two enantiomers.

While the bulk crystal structure and the DFT results already make a strong case for the presence of chiral surfaces, it is important to provide experimental validation and thereby exclude a symmetry-inducing surface reconstruction. The study of insulating mineral surfaces turned out to be a challenge with established charge-based experimental methods. Instead, the measurement of interaction forces by atomic force microscopy enables imaging of insulator surfaces including carbonates,²¹ yet, the complexity of tip–surface interactions for dolomite(104) turned out to complicate data interpretation.²² Data quality on insulator surfaces has only recently been significantly elevated by the use of Cu oxide²³ or carbon monoxide (CO)^{24,25} functionalized tips that enabled unambiguous unraveling of the atomic structures.

We succeeded in imaging the atomic structures of the two opposing (104) and ($\bar{1}0\bar{4}$) sites of the same dolomite crystal. Key for the measurements is the usage of CO-terminated tips that were precharacterized on a Ag(111) surface mounted adjacent to the dolomite crystal²⁶ (see SI section D). The $[42\bar{1}]$ and $[\bar{4}21]$ directions are determined macroscopically from the birefringence property²⁷ and images are aligned such that the respective direction points upward in all image data. The image data in Figure 1(d,e) resemble a pattern consisting of pores enclosed by a bright lattice structure. One pore (marked by a solid circle) is spatially larger than the other

(marked by an open circle). Based on the smaller ionic radius of Mg^{2+} compared to Ca^{2+} and considering the findings on calcite(104) where the bright lattice intersections are roughly located at the position of the O_{top} atoms of the carbonate groups,²⁴ we can tentatively assign the large (small) pore to the Ca (Mg) sites and center the unit cell outline vertices on Ca as indicated. Most intriguingly, the chiral character of the surfaces is directly evident: the pores express an asymmetric, pentagonal shape. The shapes marked in the images of the two crystal sides in Figure 1(d,e) by dashed white lines are clearly two enantiomers as they are nonsuperimposable mirror images of each other. At this point, we exclude a chiral center at the tip causing this difference due to our careful tip preparation protocol and the precharacterization by inverse imaging of $\text{CO}/\text{Ag}(111)$, see SI section D. Running measurements at 5 K while compensating for vertical drift²⁸ also excludes drift and creep artifacts as the source for the observed asymmetry. By imaging many different surface sites on each crystal side, we can further exclude the formation of different enantiomeric domains on a polycrystalline sample. Finally, the extended image series in SI section E highlight the presence of chirality across a large tip–sample distance range.

To precisely assign the different surface sites in the NC-AFM data, we perform image simulations for the DFT models in Figure 1(b,c), reproduced in Figure 1(f,g) with a partial overlay of the atomic geometry. There is clearly an excellent agreement between experimental and simulated data, reproducing all features. Further, the large (small) pore can unambiguously be assigned to the calcium (magnesium) site, while the bright lattice structure evolves from the interaction between the CO tip and the carbonate groups, in particular with the topmost oxygen atoms O_{top} . We note that we did not find any evidence in the experimental data and DFT models for a surface reconstruction, i.e. for a (2×1) reconstruction as has been observed and explained before on the related calcite(104) surface.^{24,29}

Next, the properties of intrinsically achiral water molecules in contact with a dolomite surface enantiomer in light of surface chirality is studied. A particular focus is set on chirality transfer as has been observed before on chiral surfaces.^{30–32} Water is dosed on a $(104)^{\text{S}}$ surface of a cooled (~ 120 K) dolomite crystal with subsequent thermal equilibration at temperatures of up to ~ 175 K (the onset of water desorption is observed above 185 K in ultrahigh vacuum). Samples with coverages in the submonolayer regime up to a single water layer are investigated by CO-tip assisted NC-AFM.

Figure 2(a) presents a topography image of the dolomite(104)^S surface covered with about 0.13 ML of water; corresponding high-resolution images acquired with a CO tip at two vertical tip–sample distances differing by about 80 pm are shown in Figure 2(b,c). Single water molecules are directly identified as features present in addition to two CO molecules as well as the dolomite(104)^S surface structure; two characteristic individual water molecules are marked by white arrows. The general contrast at large tip–sample distance in Figure 2(b) of single water molecules is similar to that of the CO-tip–water interaction already known from the calcite(104)– (2×1) system.²⁵ When additionally resolving the dolomite lattice in Figure 2(c), most water molecules are found to be located on a Ca row (large pores), while a smaller fraction resides on a Mg row (small pores). Water adsorption at these two distinct positions with a preference for the Ca row is a general observation across all experiments; we therefore

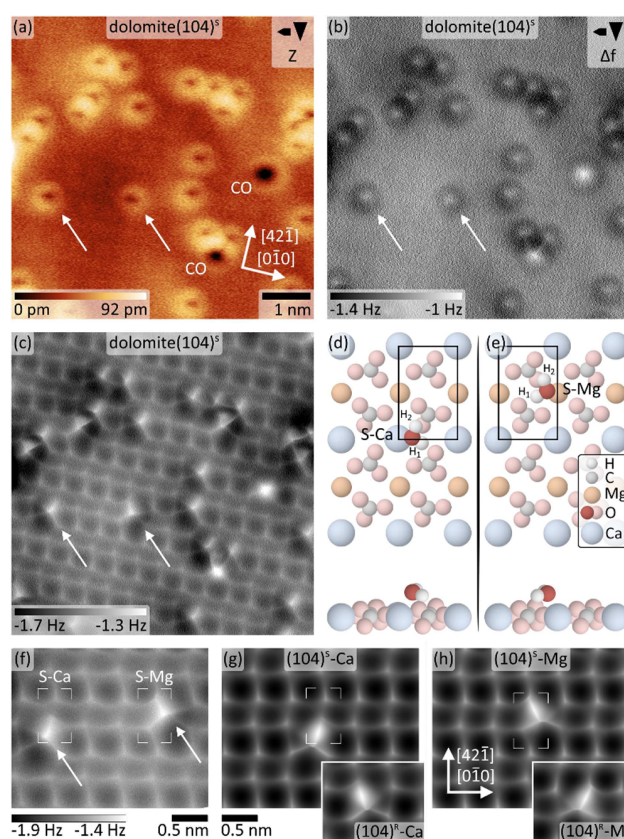


Figure 2. (a) Topography image of a dolomite(104) surface covered by water and a few CO molecules, acquired by NC-AFM at 5 K with a CO-terminated tip ($\Delta f = -1.2$ Hz and $U_{\text{sample}} = 10$ V). Single CO molecules appear as dark depressions (two CO molecules are named in panel (a)) while water molecules are imaged as bright protrusions with a smaller dark spot (two examples marked by white arrows). (b,c) High-resolution frequency shift images of the surface area shown in (a) acquired in constant-height mode with a CO-terminated tip. Images are vertically separated by 80 pm ($U_{\text{sample}} = 10$ V; vertical stability is ensured by active drift compensation²⁸). (d,e) Optimized DFT geometries of the S–Ca and S–Mg water systems. Lengths of indicated bonds are given in Table 1. (f) Close-up high-resolution image of the two S–Ca and S–Mg adsorption cases. (g,h) Simulated NC-AFM images of S–Ca and S–Mg water. Insets: mirrored images represent the R–Ca and R–Mg enantiomers.

contrast the S–Ca water with the S–Mg water in the detailed high-resolution image in Figure 2(f) and further investigate these adsorption positions on the $(104)^{\text{S}}$ surface by DFT.

Lowest energy adsorption geometries for the S–Ca and S–Mg water resulting from calculations are shown in Figure 2(d,e) with simulated NC-AFM images of these models reproduced in Figure 2(g,h). We here only focus on the $(104)^{\text{S}}$ surface enantiomer, results for the $(104)^{\text{R}}$ surface would directly follow from symmetry as shown by the mirrored insets in Figure 2(g,h). We first note an excellent agreement between the experimental and simulated NC-AFM data and thereby confirm the DFT geometries. From inspecting the models more closely, water is found to bind to the dolomite surface both via an ionic bond between the water oxygen and respective surface cation as well as via a hydrogen bond between the water H_1 and one topmost oxygen surface atom. A third bond between the second hydrogen H_2 and another surface oxygen atom is longer than what would be expected for

a hydrogen bond. The different bond lengths (see Table 1) readily justify the difference in the adsorption energy of about

Table 1. Adsorption Energies and Atomic Distances for the S–Ca and S–Mg Water Enantiomers^a

	S–Mg	S–Ca
E_{ads}	−0.98 eV	−1.03 eV
E_{ads} ref 33	−0.94 eV	−0.98 eV
$\text{O}_{\text{water}} \cdots \text{cation}$	2.12 Å	2.37 Å
$\text{H}_1 \cdots \text{O}_{\text{surf}}$	2.06 Å	1.70 Å
$\text{H}_2 \cdots \text{O}_{\text{surf}}$	2.29 Å	2.45 Å
$\text{O}_{\text{water}} \cdots \text{H}_1 \cdots \text{O}_{\text{surf}}$	3.01 Å	2.69 Å
$\text{O}_{\text{water}} \cdots \text{H}_2 \cdots \text{O}_{\text{surf}}$	3.07 Å	3.11 Å
$z_{\text{O}_{\text{water}}}$	2.02 Å	2.14 Å

^aElement identifications are introduced in Figure 2(d,e). $z_{\text{O}_{\text{water}}}$ describes the height of the O_{water} atom above the surface plane defined by the average center position of the surface cations along z .

50 meV between the two sites, in full agreement with independent recent DFT calculations.³³ The lower adsorption energy at S–Ca also confirms the experimental prevalence over S–Mg. In addition, the water oxygen–surface cation distance is by 25 pm smaller at the Mg site when compared to Ca and the H_1 (H_2) hydrogen bond of S–Mg is accordingly longer (shorter) than of S–Ca. Overall, the water oxygen atom moves 12 pm along z closer to the surface at the Mg site. The surface only undergoes minor relaxation upon adsorption of a single water molecule, see SI section F.

The strong molecule–surface interactions, different at the two cation sites, directly explains the presence of two enantiomers for each water adsorption site; the single water molecule geometry differs considerably depending at which cation site the water is adsorbed. This is a significant result as it proves chirality transfer from the dolomite(104) and ($\bar{1}04$) mineral surfaces to the achiral water molecule, leading to chiral signatures in the adsorption of single water molecules. This enantiospecific adsorption is a consequence of the strong interaction between surface and molecule; a templating effect is at work with the surface dictating the adsorption positions of the single water molecules. This is in full agreement with earlier findings of chirality transfer from chiral (mostly metallic) substrates to achiral molecules.^{30,31}

Chirality transfer is also apparent when investigating the fully hydrated surface at 1 ML water coverage. A corresponding constant-height overview image in Figure 3(a) reveals the assembly of water molecules. We start the investigation with the optimized 1 ML DFT model geometry in Figure 3(b). An analysis of the average bond lengths (see Table 2 in SI section H) highlights that the adsorption geometries at the Mg and Ca sites are rather similar to the single water cases. This finding again hints toward strong molecule–surface interactions while the water–water forces are small—supporting the templating effect and, thereby, the persistent presence of the two enantiomers. An analysis of the atomic shifts (see SI section G) between the pristine and water-covered dolomite(104) surface models further reveals that the surface atoms react with shifts of up to 25 pm to the adsorption of a full water monolayer.

While these results already hint toward chirality transfer also into the full monolayer, we give a clear answer to this central question by performing both a structural analysis of the water layer DFT geometry and a symmetry analysis of the

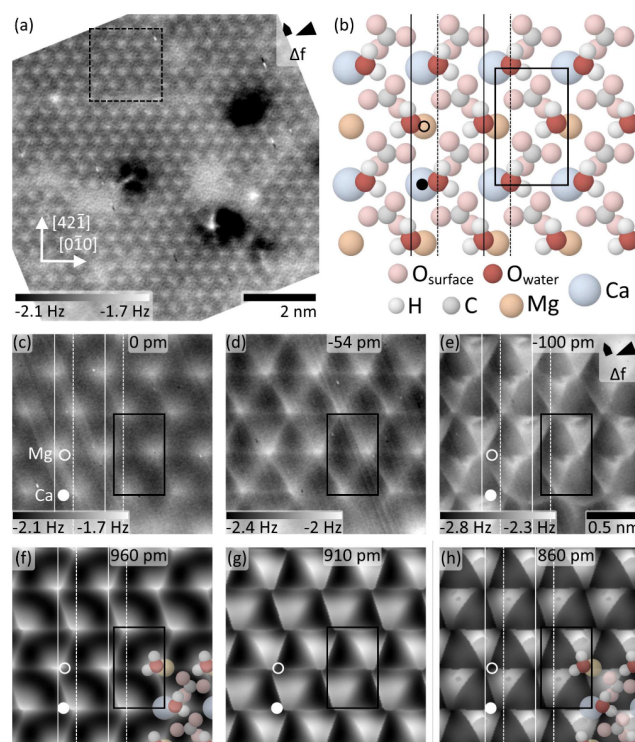


Figure 3. (a) Large-scale constant-height image (acquired at large tip–sample distance) of the dolomite(104)^S surface covered with a single layer of water molecules. (b) DFT-optimized geometry of the dolomite(104)^S surface covered by 1 ML of water. (c–e) Series of experimental constant-height images (decreasing tip–sample distance) resolving the water structure. (f–h) Simulated images of the water structure on the dolomite(104)^S surface (tip–sample distances relative to average top Ca plane included). Cation positions in (c,e) according to comparison with model, see inset in (f,h). Vertical lines in (c–h) mark the position along the horizontal axis of the primary bright features.

experimental NC-AFM data. In case of chirality transfer, there should be a notable difference between the H_2O water layer on a dolomite(104)^S and (104)^R surface. We quantify this difference by performing an adapted continuous chirality measure^{34,35} and Hausdorff distance analysis^{36,37} on the isolated water structure from the water/dolomite(104)^S model and the mirrored variant (see SI section I, also for bare dolomite). As the DFT methodology does not express any chiral bias, confirmed by the independent calculations for the pristine surface, Figure 1(b,c), we now focus only on the (104)^S surface. We calculate the translation vector $T = (\vec{r}, \theta)$ that shifts the atomic positions of the isolated water structure onto its mirrored variant while minimizing the root-mean-square deviation (RMSD) of the atomic positions, resulting in an optimal overlap of the two structures. For the isolated water/dolomite(104)^S surface we find an optimum of RMSD ≈ 16.6 pm and a corresponding Hausdorff distance of $h \approx 26.7$ pm. This means that there is no translation-rotation vector that can superimpose the water layer onto its mirror image – the isolated water layer is chiral. Overall, the DFT results give strong evidence for chirality transfer into the first water layer and allow the unambiguous identification of the two enantiomers within the DFT model as a result of the dolomite surface chirality.

Finally, we extend the analysis to the experimental data for substantiating the theoretical conclusions. We first note that the series of high-resolution NC-AFM data acquired across a tip–sample distance range of 100 pm in Figure 3(c–e) are in excellent agreement with image simulations in Figure 3(f–h), already giving evidence for the validity of the 1 ML water/dolomite(104)^S DFT model and the associated chiral property. By superimposing the atomic structure in Figure 3(f,h), the bright regions at large tip–sample distances are identified to be positioned between the water oxygen atoms and surface cations. Bright linkers appear at intermediate tip–sample distances, Figure 3(d,g), roughly in-between the water oxygen atoms, but offside of the water–carbonate hydrogen bonds. At very close tip–sample distances, these linkers evolve into triangular shapes in Figure 3(e,h). While an analysis of the shape and contrast within these individual patterns, i.e. the different separation along the $[0\bar{1}0]$ as marked by solid and dotted vertical lines, would already highlight the chiral property, we follow a comprehensive numerical approach for analyzing the symmetry, inspired by the pg symmetry test developed before for calcite(104) data.²⁴ This test calculates the mean squared error (MSE) from adding the pixelwise deviation between the data within one (1×1) unit cell and inside the mirrored version of one surface unit cell. Resulting MSE values are shown along one coordinate (y) in Figure 4(a).

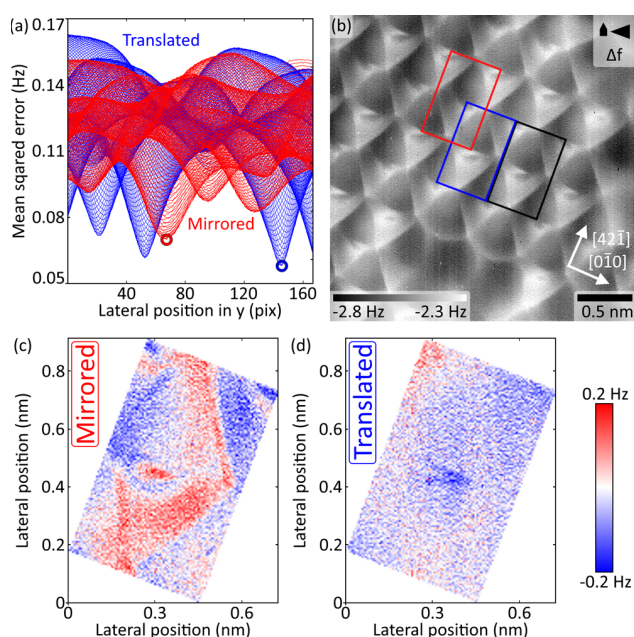


Figure 4. Symmetry analysis of the data acquired for 1 ML water on the dolomite(104)^S surface. (a) Mean squared error (MSE) for the mirrored (red) and translated (blue) variants. Minima are marked by circles. (b) Experimental data with markers for the reference unit cell in black as well as the optimum mirrored (red) and translated (blue) unit cell. (c,d) Pixel-wise deviation for the (c) mirrored and (d) translated unit cell.

Due to the surface structure, mirror axes not parallel to the $[42\bar{1}]$ direction can directly be excluded. In turn, we allow ranges along the x and y axes larger than one unit cell to find the optimum in positioning the mirrored unit cell. The minimum MSE for the mirrored unit cell marked by a red circle in Figure 4(a) corresponds to the red cell marked in Figure 4(b). To take account of experimental noise that will

fundamentally inhibit an excellent overlap with zero MSE, we compare the numbers from this chirality test with the case of a translated unit cell. Data in Figure 4(a) (blue curves) directly show a smaller MSE (blue circle) at optimum shift for the unit cell placed nearby the original one (blue rectangle in Figure 4(b)). The pixel-wise comparison within the single unit cell for these two cases directly substantiates the difference between these values: There is a strong contrast along the triangular structures in Figure 4(c) for the mirrored version, while the translated cell basically expresses noise.³⁸ Further details on the method are given in the SI section J. To conclude, the numerical analysis did not reveal a mirror symmetry in the experimental data. Instead, the water monolayer follows the enantiomeric surface structure and, thus, expresses chirality.

In summary, we deliver clear proof from experiments and simulations for the presence of two enantiomers on opposite $\{104\}$ faces of ordered dolomite and thereby confirm the expectation from the bulk crystal structure. Both enantiomers are imaged by NC-AFM using CO-terminated tips and their structures are confirmed via DFT simulations. Due to the strong molecule–surface interactions, chirality is transferred into the local structure of single water molecules as well as into the assembly of a full water monolayer, ultimately causing an achiral molecule to exhibit handedness. The energetic difference of 50 meV between water adsorption at the Ca and Mg sites suggests a measurable effect also at room temperature.

The presence of chiral dolomite surfaces has widespread implications and is of particular importance in the context of organogenic models for dolomite growth. As a consequence of chiral selectivity in biomolecules on earth, extracellular polymeric substances will consist of only one enantiomer of amino acids that is expected to favor one enantiomer of the dolomite(104) faces – either directly due to different adsorption properties of enantiomeric molecules on dolomite, or indirectly via transfer by chiral water layers. In fact, strong evidence has been given for chiral water structures in the vicinity of an antiparallel β -sheet protein,¹³ in the minor grooves of a DNA strand,¹⁴ and within assembled sugar complexes.¹⁵ With a homochiral imprint on water structures, chiral selectivity effects are expected when homochiral water is in contact with either face of dolomite. It will be the task of future studies to both quantify the extent of the surface-induced chiral imprint into higher water layers and to determine the presence of a direct or indirect chirality transfer between dolomite surfaces and biomolecules. In particular amino acids^{39,40} could contribute to the growth of dolomite, hinting towards links between chirality of dolomite{104} and the origin of homochirality of life.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.6c03227>.

Methods, nomenclature, symmetry analysis, tip characterization, constant-height image series, atomic shifts of single water molecules, atomic shifts of water monolayer, comparison of single water and full layer distances, quantification of chirality, and symmetry testing on experimental data (PDF)

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Author Contributions

¶P.L. and J.H. contributed equally. P.L. and J.Heggemann performed the experiments and analyzed the experimental data. J.Huang and A.S.F. performed the calculations. P.R. conceived the project, supported the experiments, contributed to the data analysis, and wrote the first manuscript draft. All authors discussed the results and contributed to finalising the manuscript.

Notes

The authors declare no competing financial interest.

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Supporting information: Chiral water structures on the dolomite(104)-S surface enantiomer

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A. METHODS

Sample Preparation. The experiments are performed using a double sample holder [1] supporting both a metal surface (Ag(111)/mica from Georg Albert PVD, Silz, Germany) for tip preparation and characterization as well as a dolomite crystal (SurfaceNet GmbH, Rheine, Germany) for the high-resolution NC-AFM experiments. Prior to the preparation of atomically clean surfaces, the sample holder is degassed at about 380 °C for 90 minutes.

Ag(111) surfaces are prepared by repeated sputtering (Ar^+ at about 2×10^{-6} mbar for 15 min yielding a drain current of about $2 \mu\text{A}$) and annealing (380 °C for 15 min) cycles. The dolomite crystal surfaces are prepared by cleaving the crystal with a sharp blade in vacuum [2]. Subsequently, the sample holder is heated up to 160 °C for 1 h in order to reduce surface charges on the dolomite crystal. Prior to mounting the crystals into the double sample holder, it was ensured that they are oriented such that the surfaces under investigation are from adjacent sides of the crystal.

H_2O Deposition on Dolomite(104). Water is deposited by dosing (at about 1×10^{-9} mbar) ultrapure H_2O (Millipore water for molecular Biology) into the preparation chamber from a leak valve onto the sample cooled to about 120 K. For the sub-monolayer deposition, the sample was subsequently annealed to 175 K. Prior to the dosing, the water is cleaned by more than 20 freeze-pump-thaw cycles.

CO Dosing. CO is deposited onto both samples by dosing into the cold scan head for about 26 s.

NC-AFM Experiments. Experiments are performed at 5 K with a Scienta Omicron qPlus LT gen. III AFM/STM instrument operated by a MATRX controller. A piezoelectric qPlus sensor as provided by the manufacturer is used and a bias voltage is applied on the bulk crystal surface to compensate for long-range electrostatic forces. Three steps are performed for CO tip functionalisation: First, sharp metal tips are prepared on the Ag(111) surface. Second, a CO molecule is attached to the tip via vertical manipulation. Third, the CO tip quality is ensured by reverse imaging of surface-bound CO molecules.

Data processing. All STM and NC-AFM data are processed using the open source tool Gwyddion [3]. Data are mapped to color gradients as included within each figure. A

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plane correction is applied to the topography data. Unfiltered raw data are shown unless otherwise noted.

DFT Calculations. All first-principles calculations in this work are performed using the periodic plane-wave basis VASP code [4, 5] implementing the Density Functional Theory. To accurately include van der Waals interactions for this system, we further use the Tkatchenko-Scheffler method with iterative Hirshfeld partitioning [6], shown to be best suited for ionic systems.[7, 8] Projected augmented wave potentials are used to describe the core electrons [9] with a kinetic energy cutoff of 500 eV (with PREC = accurate). Systematic k-point convergence is checked for all systems with sampling chosen according to the system size. This approach converged the total energy of all the systems to the order of 1 meV. For calculations of the dolomite surface, we use a $4 \times 4 \times 4$ supercell (320 atoms total), a vacuum gap of at least 1.5 nm, and a $5 \times 5 \times 1$ k-point grid. The upper two layers of dolomite are allowed to relax to a force of less than 0.01 eV/Å. Multiple molecular configurations were considered, and the lowest-energy structures were imaged in simulations for comparison to experiment.

NC-AFM Image Simulations. High-resolution AFM simulations for a CO-terminated tip are performed by the full density-based model (FDBM) [10, 11] as implemented by the Probe Particle Model (PPM) [12–14] with GPU acceleration. The input files for PPM simulations, including atomic structure, electrostatic potential, and electron density of the samples, are calculated from DFT through VASP. The tip-sample interaction includes three contributions: (i) Short range interaction between tip and sample is estimated by a density overlap model [11] $V_{\text{SR}} = V_0 \int [\rho^{\text{tip}}(\vec{r}) \rho^{\text{sample}}(\vec{r})]^\alpha d\vec{r}$ with parameters $V_0 = 18.0$ and $\alpha = 1.0$, represented as A_{Pauli} and B_{Pauli} in PPM. (ii) Electrostatic interaction derived from the Hartree potential and tip charge density. (iii) Van der Waals interaction treated using the DFT-D3 scheme with PBE parametrisation.

B. NOMENCLATURE

The Cahn-Ingold-Prelog (CIP) rule is established to unequivocally name the enantiomers of chiral molecules [15]. With this rule, a so-called stereocenter is determined and the groups attached to this center are assigned priorities based on the atomic numbers Z of the atoms directly attached to the stereocenter. The handedness follows from the sequence of these neighbouring atoms, in decreasing order of the atomic number. In organic molecules, the most common stereocenter is a carbon atom with at least three different attached groups. Based on this information, we define that the carbon atoms of the carbonate groups serve as starting point for our nomenclature definition.

Every carbon atom is bonded to three oxygen atoms, at first these oxygens ($Z_{\text{O}} = 8$) would have the same priority when considering the CIP-rule. However, the next neighbouring atoms for the oxygens are different: For example, the next neighbour of O_{top} of the lower carbonate group in the dolomite(104) unit cell in Fig. 1(b) of the main text is a Mg ($Z_{\text{Mg}} = 12$) atom, the next neighbours of O_{middle} are Ca ($Z_{\text{Ca}} = 20$) and Mg atoms, and the next neighbour of O_{bottom} is a Ca atom. Thus, based on the atomic numbers of the neighbouring atoms, O_{bottom} is assigned priority 1, O_{middle} priority 2, and O_{top} priority 3. The handedness finally follows from the in-plane orientation of rotation when going from priority 1 to priority 3. For the example in Fig. 1(b) of the main text, a counter-clockwise rotation is identified that defines this surface enantiomer as dolomite(104)^S. The priorities for each carbonate group on both surfaces is presented in Table S1, showing that this chirality definition indeed results in an unequivocal identification.

Based on this definition, we can unambiguously describe the dolomite(104) and dolomite($\bar{1}0\bar{4}$) surfaces as dolomite(104)^S and dolomite(104)^R, initially based on the CIP-rule for the naming of chiral molecules.

(104) surface		atom	priority	rotation
lower carbonate group	O _{top}	3		
	O _{middle}	2		S
	O _{bottom}	1		
upper carbonate group	O _{top}	1		
	O _{middle}	2		S
	O _{bottom}	3		
($\bar{1}04$) surface				
lower carbonate group	O _{top}	3		
	O _{middle}	2		R
	O _{bottom}	1		
upper carbonate group	O _{top}	1		
	O _{middle}	2		R
	O _{bottom}	3		

TABLE S1. Priorities of the oxygen atoms of the lower and upper carbonate groups in the unit cells of both dolomite surfaces.

C. SYMMETRY ANALYSIS

Fig. S1 presents the result of a numerical comparison of the atomic coordinates between the two DFT-optimised surface enantiomers. Resulting minor shifts are within the DFT accuracy.

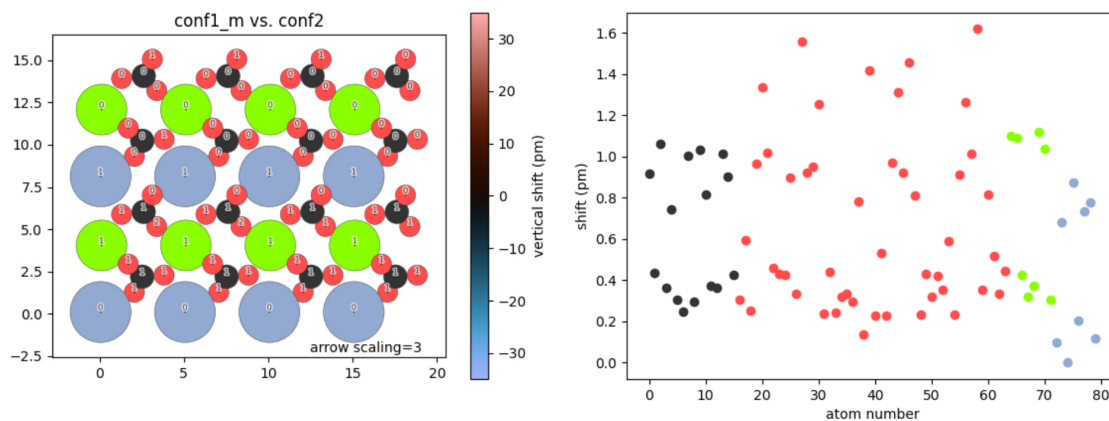


FIG. S1. (a) Top layer of the DFT-calculated dolomite{104} surfaces. Numbers denote the absolute shift between the $(104)^S$ and the mirrored $(104)^R$ surface models. (b) Absolute shifts for all atoms in the top layer. Colours are element-specific and identical to those in (a).

D. TIP CHARACTERISATION

CO-terminated tips were characterised before performing the constant-height experiments by means of reverse imaging of a surface bound CO on the Ag(111) surface. The respective STM and frequency shift images (acquired in STM feedback) are shown in Fig. S2 top and bottom row, respectively. Note that the increased noise level is due to fast scan speeds that became necessary to avoid molecular manipulation. Due to the mesoscopic tip characteristics, the effective tip-sample distance is closer in Fig. S2(b,f) than in the other cases, also explaining the stronger contrast in the NC-AFM data (lower panel). As explained in Ref. [16], CO molecules deflect under these conditions. All images clearly express symmetric images of CO molecules, providing evidence for symmetric CO tips.

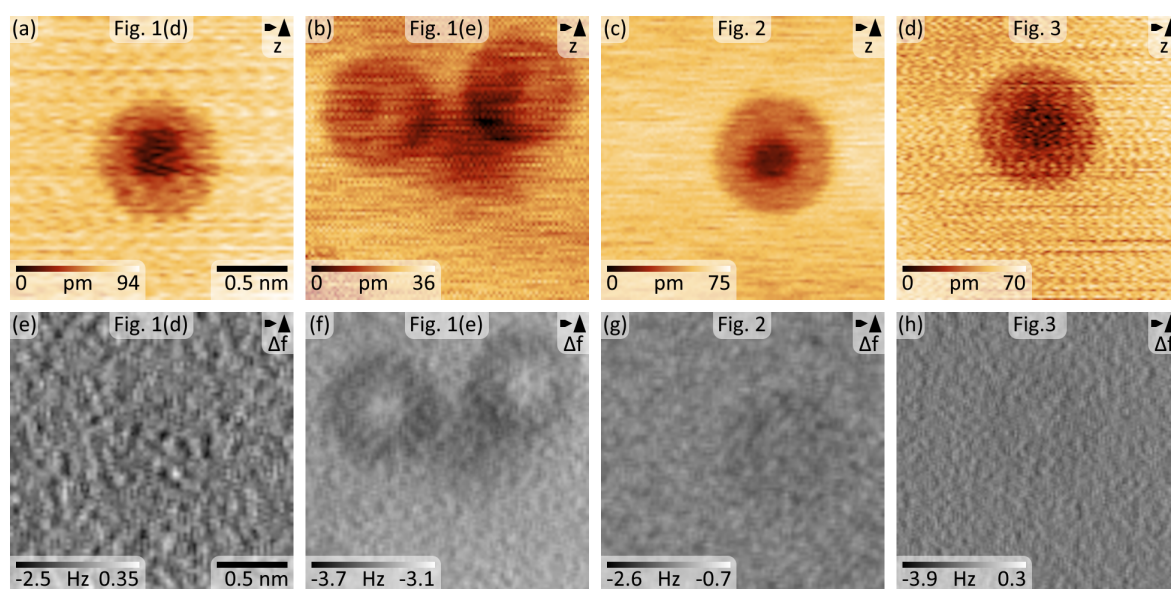


FIG. S2. (a-d) STM images and (e-h) corresponding frequency shift data of surface-bound CO on Ag(111) after CO-tip functionalisation for the purpose of tip characterisation. The labels at the top refer to the images in the main text. Plane- and line-wise correction applied to topography image data. Noise in the frequency shift data is reduced by using a 3 pixel averaging filter.

E. CONSTANT-HEIGHT IMAGE SERIES

Data in Fig. 1 of the main text are extracted from extended constant-height series acquired on the two surfaces; further data acquired at other tip-sample distances are presented in Fig. S3. Relative image heights are determined from comparing frequency-shift values from specific image sites with values acquired via $\Delta f(z)$ data acquisition. Notably, chirality is observed at all tip-sample distances.

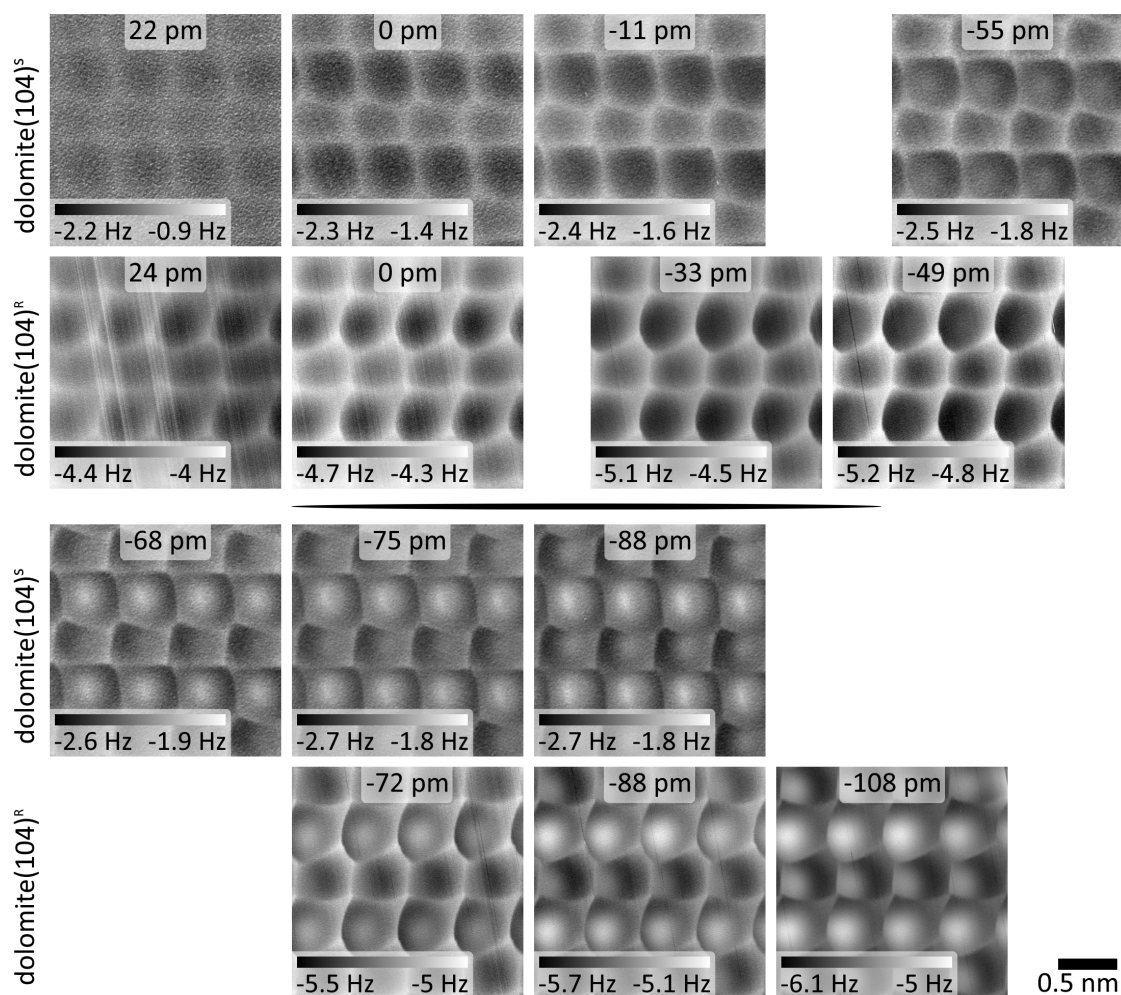


FIG. S3. Frequency shift images acquired in constant-height mode on the two enantiomeric sample surfaces across a tip-sample distance regime of 110 pm (132 pm) on the dolomite(104)^S (dolomite(104)^R) surface.

F. ATOMIC SHIFTS: SINGLE WATER MOLECULES

An analysis of the atomic shifts upon adsorption of a single water molecule at the Ca and Mg site on a $(104)^S$ dolomite surface is shown in Fig. S4.

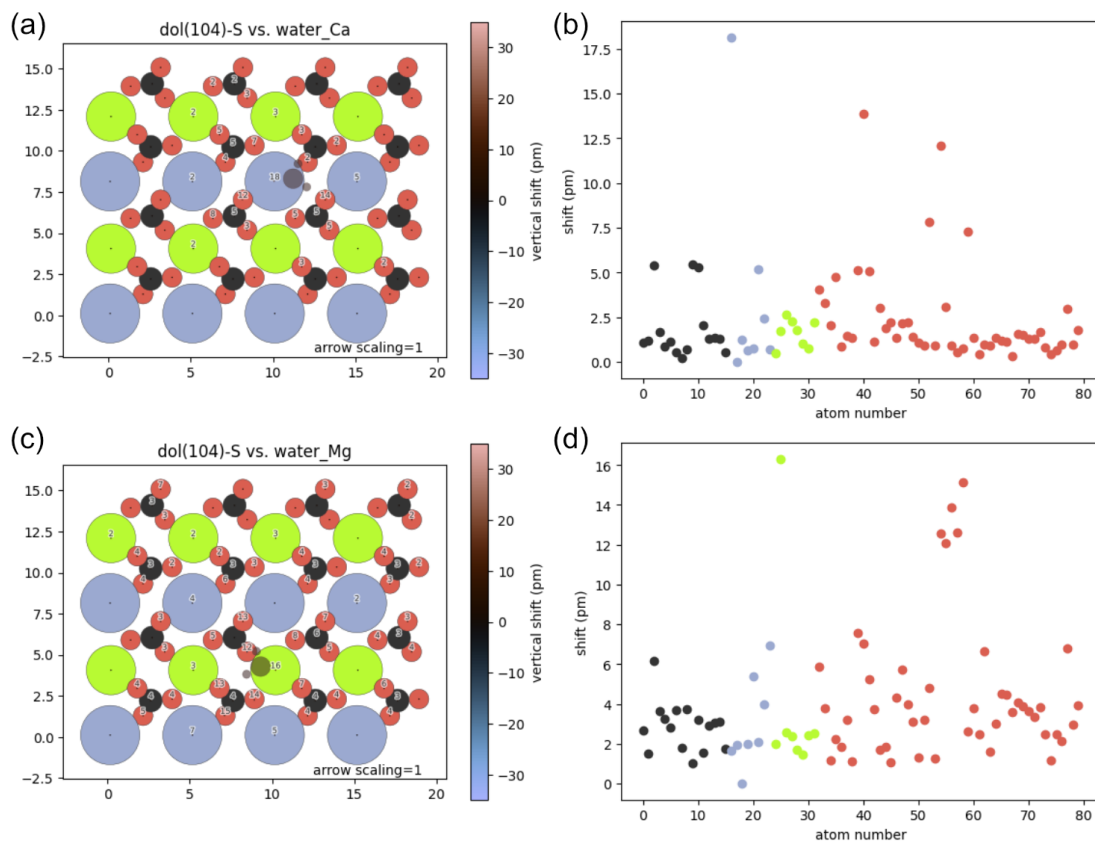


FIG. S4. (a,c) Top layer of the DFT-calculated dolomite(104)^S and H₂O/dolomite(104)^S surfaces for water adsorption at the Ca and Mg surface site. Numbers denote the absolute shift between the (104)^S and the respective H₂O/dolomite(104)^S surfaces surface models. (b,d) Absolute shifts for all atoms in the top layer. Colours are element-specific and identical to those in (a and c).

G. ATOMIC SHIFTS: WATER MONOLAYER

An analysis of the surface atomic shifts upon adsorption of a full water monolayer on a $(104)^S$ dolomite surface is shown in Fig. S5. All carbonate groups undergo a relaxation with magnitudes slightly larger than the single-water cases (see Fig. S4).

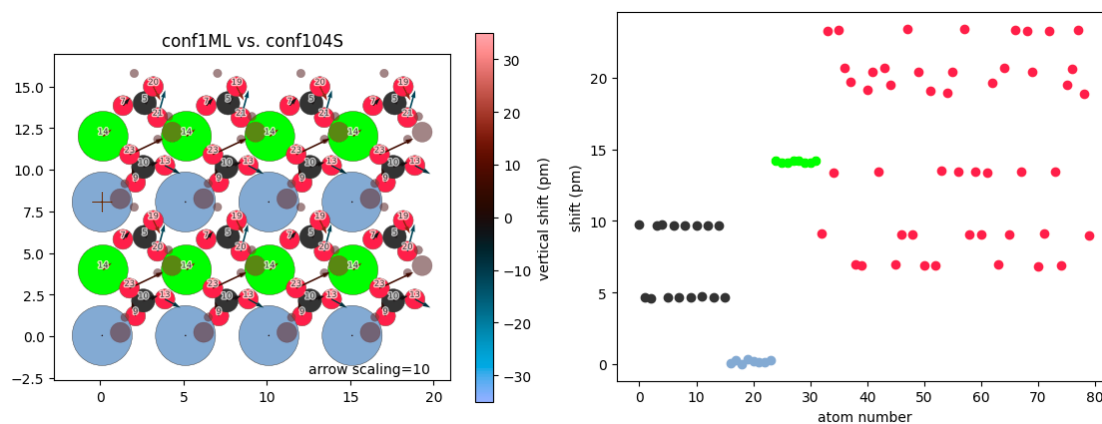


FIG. S5. (a) Top layer of the DFT-calculated dolomite $(104)^S$ and $\text{H}_2\text{O}/\text{dolomite}(104)^S$ surface. Numbers denote the absolute shift between the $(104)^S$ and the respective $\text{H}_2\text{O}/\text{dolomite}(104)^S$ surface models. (b) Absolute shifts for all atoms in the top layer. Colours are element-specific and identical to those in (a).

H. COMPARISON OF SINGLE WATER AND FULL LAYER DISTANCES

Table S2 compares the bond distances for the single water models and the full water layer. As there are only minor differences within the single/full pairs, it can be concluded that the adsorption geometries at both surface sites are rather similar when comparing a single water molecule with adsorption of a full layer.

	H ₂ O–Mg single	H ₂ O–Mg full	H ₂ O–Ca single	H ₂ O–Ca full
O _{water} ···cation	2.12 Å	2.12 Å	2.37 Å	2.39 Å
H ₁ ···O _{surf}	2.06 Å	2.00 Å	1.70 Å	1.71 Å
H ₂ ···O _{surf}	2.29 Å	2.25 Å	2.45 Å	2.48 Å
O _{water} ···H ₁ ···O _{surf}	3.01 Å	2.95 Å	2.69 Å	2.69 Å
O _{water} ···H ₂ ···O _{surf}	3.07 Å	3.03 Å	3.11 Å	3.17 Å
z _{O_{water}}	2.02 Å	1.95 Å	2.14 Å	2.07 Å

TABLE S2. Interatomic distances for single water molecules and water molecules from the full water layer on both the Mg and the Ca surface site.

I. QUANTIFICATION OF CHIRALITY

As a structure can either be chiral or achiral, chirality itself is in general a binary property. However, from the necessity to quantify chirality when comparing different chiral objects, a quantification has been proposed [17]. In order to quantitatively describe the chirality of the dolomite(104) and water/dolomite(104) system, an analysis based on an adapted continuous chirality measure (CCM) [18, 19] and Hausdorff distance approach [20, 21] is performed. For a given chiral structure, CCM determines the closest related achiral structure by minimizing the root mean square deviation (RMSD) of the atomic positions between the original chiral and the deduced achiral structure. The RMSD becomes larger if the atomic positions have to be shifted further. In this case, the chirality of the structure in question can be seen as “larger” as its atoms are further away from any achiral structure. The Hausdorff distance h on the other side is defined as the maximum deviation between a structure and its mirror image after optimally superimposing both structures by means of translation and rotation. Here, the optimal superimposition is reached when the Hausdorff distance is minimized.

For the case of a chiral surface, these two methods are combined. In order to find the optimum overlap of the two surface enantiomers, a translation vector $T = (\vec{r}, \theta)$ is defined that shifts enantiomer R onto enantiomer S while minimizing the RMSD between the element-specific atomic positions of the enantiomer DFT geometries. Note that this does neither strictly resemble the CCM nor the Hausdorff distance approach as neither the closest related achiral structure is searched, nor is the Hausdorff distance minimised.

First, results on the bare dolomite surface are presented. The analysis reveals an optimum overlap with RMSD ≈ 77.7 pm and a Hausdorff distance of $h \approx 110.5$ pm. Fig. S6(a) shows the differences in atomic positions after optimally superimposing the dolomite $(104)^S$ and $(104)^R$ surface geometries. The atomic shift of each individual atom is reproduced in (b). It can clearly be seen that the largest distances in the atomic positions of roughly 100 pm appear for the oxygen atoms, while the cations as well as the carbons all have smaller distances of below 20 pm. This can directly be understood from the fact that the chirality of the dolomite $\{104\}$ surface mainly originates from the different orientations of the carbonate groups within the surface unit cell.

The values for the isolated (surface removed in the model) water monolayer structure are determined as RMSD ≈ 16.6 pm and $h \approx 26.7$ pm. This can be understood as the water

structure being less chiral than the bare dolomite surface. Similar to the bare surface, the differences in atomic positions and the individual atomic shifts are shown in Fig. S7(a,b).

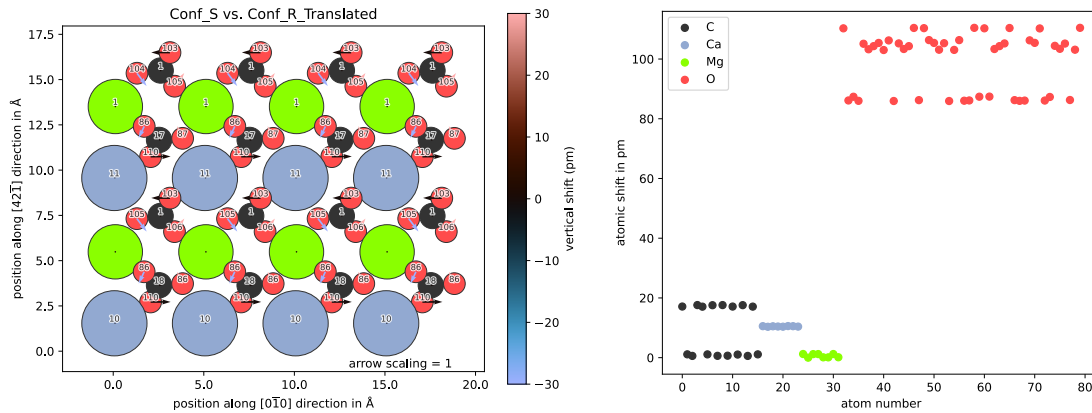


FIG. S6. (a) Comparing the top layers of the DFT-calculated dolomite(104)^S and (104)^R surfaces. Numbers denote the absolute shifts between the atoms of the (104)^S and (104)^R surface models after optimal superimposition. (b) Absolute distances between atoms after optimal superimposition. Colours are element-specific and identical to those in (a).

In order to check for any numerical bias in the determination of the translation vector and the corresponding RMSD and Hausdorff distance values, we also performed the analysis with two surface structure models that are known to be achiral. The first set of surface structures consists of the mirrored dolomite (104)^S geometry and the dolomite (104)^R geometry. For this structure pair we find RMSD ≈ 0.7 pm and $h \approx 1.4$ pm, which is less than the noise known for DFT optimisations, meaning that achirality is confirmed. The second test is performed with the achiral surface structure calcite(104) [22] and its mirrored copy. Here, we find RMSD ≈ 0.8 pm and $h \approx 1.9$ pm, again with the DFT noise and, thus, confirming the achiral character.

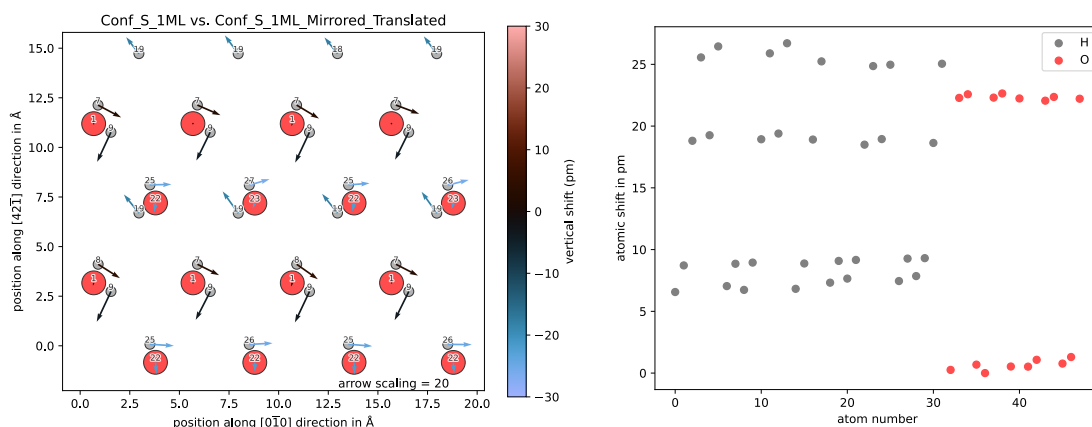


FIG. S7. (a) Comparing the top layer of the DFT-calculated $\text{H}_2\text{O}/\text{dolomite}(104)^S$ monolayer structure and its mirrored version (all dolomite surface atoms removed before analysis). Numbers denote the absolute shifts between the atoms of the surface models after optimal superimposition. (b) Absolute distances between atoms after optimal superimposition. Colours are element-specific and identical to those in (a).

J. SYMMETRY TESTING ON EXPERIMENTAL DATA

Mean squared error (MSE) data for different translation vectors mapped to the surface plane are shown in Fig. S8. Panel (a) presents data for the mirrored unit cell and panel (b) reproduces the cross-validation from the purely translated unit cell. Data are identical to data in Fig. 4 of the main text, but arranged within the surface plane according to the final translation position. The minima at the unit cell centre (mirrored variant in Fig. S8(a)) and at the unit cell vertices (translated variant in Fig. S8(b)) are clearly visible.

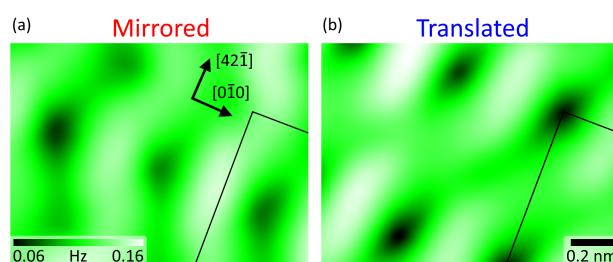


FIG. S8. Mean squared error mapped with respect to the final translation position for (a) the mirrored and (b) the original unit cell, both translated for comparing the the original data.

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